

Linear Algebra Done Right (4th Ed.) – Axler

Matrix of the Differentiation Map

Problem. Suppose $D \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}), \mathcal{P}_2(\mathbb{R}))$ is the differentiation map defined by $Dp = p'$. Find a basis of $\mathcal{P}_3(\mathbb{R})$ and a basis of $\mathcal{P}_2(\mathbb{R})$ such that the matrix of D with respect to these bases is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Solution. Recall that if $T \in \mathcal{L}(V, W)$, and $v_1, \dots, v_n$ and $w_1, \dots, w_m$ are bases of V and W respectively, then the k -th column of $\mathcal{M}(T)$ records the coordinates of Tv_k in terms of the w_j basis. In particular, the given matrix requires

$$Dv_k = w_k \quad \text{for } k = 1, 2, 3, \quad \text{and} \quad Dv_4 = 0.$$

Choosing a basis for $\mathcal{P}_3(\mathbb{R})$. Since $\dim \mathcal{P}_3(\mathbb{R}) = 4$, we need four linearly independent polynomials. We take the standard monomial basis:

$$v_1 = x^3, \quad v_2 = x^2, \quad v_3 = x, \quad v_4 = 1.$$

Computing the derivative of each basis vector gives

$$Dv_1 = 3x^2, \quad Dv_2 = 2x, \quad Dv_3 = 1, \quad Dv_4 = 0.$$

Choosing a basis for $\mathcal{P}_2(\mathbb{R})$. We need the basis vectors w_1, w_2, w_3 of $\mathcal{P}_2(\mathbb{R})$ to satisfy $w_k = Dv_k$. The computations above dictate the choice:

$$w_1 = 3x^2, \quad w_2 = 2x, \quad w_3 = 1.$$

Since $\dim \mathcal{P}_2(\mathbb{R}) = 3$ and these three polynomials are clearly linearly independent (they have distinct degrees), they form a basis of $\mathcal{P}_2(\mathbb{R})$.

Verification. With these choices, the action of D on each basis vector of $\mathcal{P}_3(\mathbb{R})$ is:

$$Dv_1 = 3x^2 = 1 \cdot w_1 + 0 \cdot w_2 + 0 \cdot w_3,$$

$$Dv_2 = 2x = 0 \cdot w_1 + 1 \cdot w_2 + 0 \cdot w_3,$$

$$Dv_3 = 1 = 0 \cdot w_1 + 0 \cdot w_2 + 1 \cdot w_3,$$

$$Dv_4 = 0 = 0 \cdot w_1 + 0 \cdot w_2 + 0 \cdot w_3.$$

Reading off the coordinate columns yields exactly the desired matrix:

$$\mathcal{M}(D) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Comparison with Example 3.33. In Example 3.33, the standard monomial bases $1, x, x^2, x^3$ and $1, x, x^2$ are used for both spaces. Under those bases, the matrix of D contains the scalar coefficients $1, 2, 3$ along its super-diagonal, reflecting the factors produced by differentiation. Here, by choosing $w_k = Dv_k$ for each k , we absorb those scalar factors directly into the basis vectors of $\mathcal{P}_2(\mathbb{R})$, yielding the cleaner identity-like matrix above.